

Haokai Ding

Shanghai, China

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Summary

Robotics researcher focusing on underactuated grasping, idle-stroke and Peaucellier-based linkages, and contact-rich aerial manipulation. Building hardware prototypes and analytical models that bridge mechanism design with field-deployable manipulation. Incoming M.S. student in Robotics at MBZUAI.

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

M.S. IN ROBOTICS (INCOMING)

- Advisor: Prof. Xingxing Zuo.

Abu Dhabi, UAE

Fall 2026

Tsinghua University | Shenzhen X-Institute

TSIEN EXCELLENCE IN ENGINEERING PROGRAM (TEEP) – X SCHOLAR

- Joint training program in underactuated robotics and mechanism design.

Shenzhen, China

Sep. 2023 – Jul. 2026

Shenzhen Technology University – Future Technology School

B.ENG. IN ELECTRONIC SCIENCE AND TECHNOLOGY

- GPA: 85.08/100. Research focus: robotics and mechanism design. Top 10 Outstanding Graduate.

Shenzhen, China

Sep. 2022 – Jul. 2026

Research Experience

State Key Lab of Mechanical Systems & Vibration, Shanghai Jiao Tong University

VISITING STUDENT – UAV PERCHING AND AERIAL MANIPULATION

- Advisor: Prof. Wei Dong.
- Working on UAV perching and contact-rich aerial manipulation.

Shanghai, China

Feb. 2026 – Present

Shenzhen X-Institute, Tsinghua University

UNDERGRADUATE RESEARCHER – UNDERACTUATED ROBOTIC GRIPPERS

- Designed and tested underactuated grippers based on semi-Peaucellier linkages.
- Developed idle-stroke and delay-triggering mechanisms; validated prototypes via simulation and physical experiments.

Shenzhen, China

2023 – 2026

Publications

† Co-first author * Corresponding author

SELECTED CONFERENCE PAPERS (ROBOTICS)

[P1] A Novel Gripper with Semi-Peaucellier Linkage and Idle-Stroke Mechanism for Linear Pinching and Self-Adaptive Grasping

H. DING, W. ZHANG*

IEEE/RSJ IROS 2025

[P2] Semi-Peaucellier Linkage and Differential Mechanism for Linear Pinching and Self-Adaptive Grasping

H. DING, Z. CHEN, T. YANG*, W. ZHANG*

IEEE CASE 2025

[P3] BioCrest – A Flexible Adaptive Robotic Hand Based on Idle-Stroke Mechanism

H. DING, X. ZHONG, W. HUANG, T. YANG*, W. ZHANG*

IEEE ROBIO 2025

[P4] GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation

H. DING, H. LUAN, T. YANG*, W. ZHANG*

IEEE ICCMA 2025

[P5] Peaucellier Gripper: A Novel Underactuated Gripper for Linear Pinching and Self-Adaptive Grasp

H. DING, J. FAN, Z. ZHU, Y. ZHAO, K. CHEN*, W. ZHANG*

IEEE ICARM 2025

[P6] An Underactuated Gripper with Grasshopper-Inspired Linkages and Delay Triggering for Pinching and Adaptive Grasping

H. DING†, Y. CHEN†, D. JIA, W. ZHANG*

IEEE ICIA 2025

[P7] An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping

Y. CHEN, H. LUAN, **H. DING**, W. ZHANG*

IEEE RAAI 2025

OTHER PUBLICATIONS

[P8] Recent Progress on the Research of 3D Printing in Aqueous Zinc-Ion Batteries

Y. LIU[†], **H. DING**[†], H. CHEN, H. GAO, J. YU, F. MO, N. WANG*

Polymers (MDPI), 2025

[P9] Rapid Analysis of Heavy-Metal Element Adsorption by SCG Based on LIBS Technology

H. XIE, L. YOU, X. FANG, L. LI, **H. DING**, Z. ZHOU, G. ZHANG, D. ZHANG

E3S Web of Conferences, 2025

Honors & Awards

2026	Top 10 Outstanding Graduate , Shenzhen Technology University	Shenzhen, China
2025	Undergraduate Champion , ASME Student Mechanism & Robot Design Competition	USA
2025	National Scholarship , Ministry of Education of China	China
2024	President's Award , Shenzhen Technology University	Shenzhen, China
2023	Gold Prize , 9th China "Internet+" Innovation Competition (Guangdong Division)	Guangdong, China
2023	Second Prize , 17th "Challenge Cup" Extracurricular Science Competition (Guangdong Division)	Guangdong, China

Professional Service

2025	Conference Reviewer , IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
2025	Session Chair , IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
2025	Conference Reviewer , IEEE International Conference on Automation Science and Engineering (CASE)